NATIONAL INSTITUTE OF TECHNOLOGY
KURUKSHETRA

Mid-Term Examination I: *Feb'2025*
Subject: *Data Structures [CSPC100/ ITPC100/ AIPC100]*
Branches: *CSE, IT, AI&ML (Regular/ Reappear)*
NOTE: 1. *The question paper contains* ***SEVEN*** *questions.*
2. *All questions are compulsory.*
3. *Assume suitable data, if missing.*

MM: *15*
Duration: *50 mins*

Q1. Find the time complexity of the following codes (in terms of Big-O notation):-

(i)

```
void my(int n){
    int i;
    for(i=1; i<=n; i++)
        for(i=1; i<=n; i++)
            printf("NIT   ");
}
```

(ii)

```
void my(int n){
    int i=1, s=1;
    while(s<=n){
        i++;
        s=s+i;
        printf("KKR ");
    }
}
```

Q2. Find a recurrence for the following recursive algorithm and solve the recurrence to find the time complexity in terms of Big-O notation.

```
f(int n){
        if(n<=1) return;
        for(int i=1; i<=3; i++)
                f(n/3);
}
```

Q3. What will be the output of the following C program? If there is a compile-time or run-time error(s), then give reasons for the same. // Assume that required header files are included.

```
int main() {
    char x[] = "NITKKR2025";
    char *ptr = x;
    printf("%s", ptr + ptr[4] - ptr[1]);
    return 0;
}
```

Q4. Suppose that for inputs of size n on a particular computer, insertion sort runs in $8n^2$ steps and merge sort runs in $64n \lg n$ steps. For which values of n does insertion sort beat merge sort?

Q5. In a lower triangular matrix **a**, with **n** rows, the maximum number of nonzero terms in a row **i** is **i+1**. Thus the total number of nonzero terms is

$$d = \sum_{i=0}^{n-1} (i+1) = n(n+1)/2$$

Since storing a lower triangular matrix as a 2-D array wastes space, we would like to find a way to store only the nonzero terms in the lower triangular matrix. Find an addressing formula for the elements a_{ij} so that they can be stored by rows in an array **b[d-1]**, with **a[0][0]** being stored in **b[0]**.
// Assume that the array index starts from 0.
// A lower triangular matrix is one in which all the elements above the main diagonal of a square matrix are zero.

Q6. Given that the median of *n* elements can be found in *O(n)* time. Write a recurrence for the runtime of quick sort, in which the median is selected as a pivot. Further, solve this recurrence using Master's Theorem.

Q7. In a permutation $a_1, a_2, \ldots a_n$, of **n** distinct integers, an inversion is a pair (a_i, a_j) such that $i < j$ and $a_i > a_j$.

(i) List the five inversions of the array < 2, 3, 8, 6, 1 >.

(ii) What would be the worst-case time complexity of the Insertion Sort algorithm, if the inputs are restricted to permutations of **1, 2, … n** with at most **n** inversions?
(Select one correct option & explain your answer)

a) $O(n^{0.5})$ b) $O(n)$ c) $O(n^{1.5})$ d) $O(n^2)$ e) $O(n \log n)$

Marks Distribution

Q. 1 to 6: 2
Q. 7: 1+2